



Beijing-Dublin International College



SEMESTER II FINAL EXAMINATION – 2019/2020

BDIC1030J & BDIC1026J Maths 2 (Advanced Mathematics)

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Time Allowed: 90 minutes

Instructions for Candidates

Answer ALL questions. The marks that each question carry is written as shown.

BJUT Student ID: _____

UCD Student ID: _____

I have read and clearly understand the Examination Rules of both Beijing University of Technology and University College Dublin. I am aware of the Punishment for Violating the Rules of Beijing University of Technology and/or University College Dublin. I hereby promise to abide by the relevant rules and regulations by not giving or receiving any help during the exam. If caught violating the rules, I accept the punishment thereof.

Honesty Pledge: _____ **(Signature)**

Instructions for Invigilators

Non-programmable calculators are permitted. NO dictionaries are permitted.
No rough-work paper is to be provided for candidates.

NOTE: Answer **ALL** questions.

Time allowed is **90** minutes.

The exam paper has **2** sections on **6** pages, with a full score of 100 marks.

You are required to use only the provided **Examination Book** for answers.

SECTION A — Brief Answer Questions

This section is worth a total of **70** marks, with each question worth **5** marks.

1. Compute

$$\int \frac{x^2 - 1}{x^4 + 1} dx = \text{_____}.$$

2. Compute

$$\int x \tan^2 x dx = \text{_____}.$$

3. Compute

$$\int \frac{\sin \ln(x + \sqrt{1 + x^2})}{\sqrt{1 + x^2}} dx = \text{_____}.$$

4. Compute

$$\int \sqrt{\frac{x + 1}{x - 1}} dx = \text{_____}.$$

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5. Compute

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left\{ \frac{\arctan^3 x}{\sqrt{1+x^2}} - \cos^{2019} x \right\} dx = \underline{\hspace{2cm}}.$$

6. Evaluate the definite integral

$$\int_0^{\frac{\pi}{2}} \frac{\cos^{2019} x}{\cos^{2019} x + \sin^{2019} x} dx = \underline{\hspace{2cm}}.$$

7. Evaluate the definite integral

$$\int_0^2 x^{2019} \sqrt{2-x} dx = \underline{\hspace{2cm}}.$$

8. Determine if the following improper integral is convergent or not:

$$\int_e^{+\infty} \frac{1}{x \ln^3 x} dx.$$

If this integral is convergent, find its value: $\underline{\hspace{2cm}}$.

9. Given a function $f(x) = \frac{\sqrt{1-x^2}}{\pi} - (x^2+1) \int_0^1 f(x) dx$, find the definite integral $\int_0^2 f(x) dx$.

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10. Evaluate the definite integral

$$\int_0^2 x e^{|x-1|} dx = \underline{\hspace{2cm}}.$$

11. Try to estimate the following limit by making use of the definition of definite integral:

$$\lim_{n \rightarrow \infty} \frac{\pi}{n+1} \sum_{k=1}^n \sin \frac{k\pi}{n}.$$

12. Given the conditions

$$y' = \sqrt{y} \cos^2 \sqrt{y}, \quad y(2) = \frac{\pi^2}{16},$$

try to determine the function $y(x) = \underline{\hspace{2cm}}$.

13. Given an ODE $y'' - 6y' + 9y = 0$, the general solution of this ODE can be expressed as
 $\underline{\hspace{2cm}}.$

14. Judge the type of the following ODE:

$$(x^3 + y^3)dx - (y^3 - x^2y + xy^2)dy = 0$$

It is a $\underline{\hspace{2cm}}$ (first-order/second-order), $\underline{\hspace{2cm}}$ (linear/non-linear)
 $\underline{\hspace{2cm}}$ (homogeneous/non-homogeneous) ordinary differential equation.

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SECTION B — Extended Answer Questions

Write your answers on the **Examination Book** provided.

This section is worth a total of **30** marks. The marks of each question are as shown.

- 15. (8 marks)** Find the general solution of the following ODE:

$$y'' - 3y' + 2y = xe^x.$$

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16. (16 marks) In the two-dimensional plane, there is a region, denoted by D , enclosed by x -axis and a parabola $y = 3x - x^2$.

- (a) Find the area of this region. (4 marks)
- (b) Find the volume of the solid obtained by revolving the region about the vertical line $x = -1$. (4 marks)
- (c) Determine Express the length of the circumference of the region. (4 marks)
- (d) If the region D is revolved about x -axis to generate a solid. Express the area of the surface by means of definite integral. (4 marks)

17. (6 marks) Suppose that $f(x)$ is monotonically continuously increasing on $[a, b]$. Show that

$$\int_a^b xf(x)dx \geq \frac{a+b}{2} \int_a^b f(x)dx.$$

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USEFUL FORMULAE

$$\int \frac{1}{\sqrt{x^2 + 1}} dx = \ln(x + \sqrt{x^2 + 1}) + c$$

$$\int \frac{1}{\cos^2 x} dx = \tan x + c$$

$$\int \frac{1}{\sin^2 x} dx = -\cot x + c$$

$$\int \tan x dx = -\ln |\cos x| + c$$

$$\int \cot x dx = \ln |\sin x| + c$$

$$\int \frac{1}{1 + x^2} dx = \arctan x + c$$

$$\int \frac{1}{x} dx = \ln x + c$$

$$\int \frac{1}{\sqrt{1 - x^2}} dx = \arcsin x + c$$

Glossary

Antiderivative	原函数
Boundary	边界
Circumference	周长
Coefficient	系数
Convergent	收敛的
Curve	曲线
Definite integral	定积分
Even function	偶函数
General solution	通解
Homogeneous	齐次的
Improper integral	广义积分
Odd function	奇函数
ODE (Ordinary differential equation)	常微分方程
Region	区域
Shadowed	阴影
Volume	体积
Area of a surface	表面积